Capstone Project 1 Report

"CricVision: Interactive IPL Cricket Analytics Dashboard"

Domain: Sports Analytics

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1. Introduction

The Indian Premier League (IPL) generates a large amount of cricket data across seasons, including batting, bowling, match results, player performances, and team statistics. Analysing this data manually can make it difficult to identify performance patterns, compare players, and understand team success across different seasons.

This project focuses on analysing IPL data using SQL and Power BI to transform raw cricket data into meaningful and interactive insights. The dashboard provides an easy way to explore team and player performance through season-based filters, key performance indicators, tables, and interactive visualizations.

The project is designed to provide a centralized analytical view of IPL performance, helping users understand batting and bowling trends, team standings, player achievements, and important tournament statistics.

2. Problem Statement

IPL datasets contain a large amount of detailed information, but raw data alone is difficult to interpret and analyse efficiently. Finding meaningful information about player performance, team performance, and season-wise trends from large datasets can require significant manual effort.

Traditional cricket statistics often present information as individual tables or static records, making it difficult to compare different players, teams, and seasons interactively.

Therefore, there is a need for an interactive analytics platform that can:

- Analyse large volumes of IPL data efficiently.
- Provide important player and team performance statistics.
- Compare batting and bowling performances.
- Display season-wise trends.
- Provide interactive filtering based on seasons and players.
- Present important tournament achievements such as Orange Cap, Purple Cap, Champion, and Runner-up.
- Help users understand IPL performance through clear visualizations and KPIs.

CricVision addresses these requirements by combining SQL-based data processing with an interactive Power BI dashboard.

3. Objective

The main goals of this initiative include:

- Conducting a detailed review of past IPL matches, player statistics, and performance metrics.
- Utilizing SQL to convert unstructured cricket records into organized, analytical data frameworks.
- Establishing significant key performance indicators for evaluating batting and bowling efficiency.
- Assessing the contributions of individual players across various tournament editions.
- Examining the collective success of teams and their positions in the league table.
- Pinpointing the most impactful athletes in both batting and bowling categories.
- Presenting critical tournament honors, including the Orange and Purple Cap winners.
- Implementing dynamic filters to explore data by specific seasons or individual cricketers.
- Building a user-friendly Power BI interface for comprehensive sports data exploration.
- Illustrating the practical value of advanced analytics within the sporting industry.

4. Tools and Technologies

Tool / Technology	Purpose
MySQL	Data storage, transformation, SQL queries, and creation of analytical views
SQL	Data aggregation, filtering, joins, calculations, and analytical processing
Power BI	Interactive dashboard development and data visualization
DAX	Creation of dynamic KPIs, calculated tables, and measures

Power Query	Data preparation and transformation before visualization
CSV / JSON Data	Source IPL cricket datasets
Power BI Slicers	Interactive filtering by season and player

5. Methodology

The project follows a structured data analytics workflow.

5.1. Data Collection

IPL cricket data containing match, player, batting, bowling, and team information was collected from structured cricket datasets.

The data contains detailed match-level and delivery-level information that can be used to derive player and team statistics.

5.2. Data Storage

The raw data was imported into **MySQL**, where the datasets were organized into appropriate tables.

MySQL was used as the primary database for storing and querying the IPL data.

5.3. Data Transformation

SQL queries were used to transform raw data into analytical datasets.

Operations included:

- Filtering records.
- Joining different tables.
- Aggregating player statistics.
- Calculating batting statistics.
- Calculating bowling statistics.
- Calculating team-level statistics.
- Creating SQL views for Power BI.
- Separating league-stage matches from playoff matches for the points table.

5.4. Data Modelling

The processed data was connected to Power BI and relationships were created between relevant tables.

A calculated **Dim Player** table was created by combining players from the Player Summary, Batting Stats, and Bowling Stats datasets.

This enabled common player-based filtering across the dashboard.

5.5. DAX Analysis

DAX measures were created to calculate dynamic KPIs such as:

- Total runs
- Batting average
- Strike rate
- Fours
- Sixes
- Wickets
- Bowling average
- Economy
- Bowling strike rate
- Dot balls
- Five-wicket hauls
- Matches played

5.6. Dashboard Development

Power BI was used to create three interactive dashboard pages:

- Executive Dashboard
- Batting Intelligence
- Bowling Intelligence

Interactive slicers were added to allow users to select seasons and players.

5.7. Dynamic Images

A separate player-image table was incorporated into the Power BI model.

The player image changes dynamically according to the selected player, allowing player profiles and achievement cards to display relevant images.

This creates a more engaging and user-friendly dashboard experience.

6. Data Description

The project uses IPL cricket datasets containing information about matches, players, batting performances, bowling performances, and individual deliveries.

Major Data Categories

Match Data

Contains information such as:

- Match ID
- Season
- Match date
- Venue
- City
- Team 1
- Team 2
- Toss winner
- Toss decision
- Winner
- Result
- Player of the Match

Batting Data

Contains fields such as:

- Match ID
- Season
- Batter
- Batting team
- Runs
- Balls faced
- Fours
- Sixes
- Dot balls
- Strike rate
- Dismissal information

Bowling Data

Contains fields such as:

- Match ID
- Season

- Bowler
- Bowling team
- Legal balls
- Overs
- Runs conceded
- Dot balls
- Fours conceded
- Sixes conceded
- Wickets
- Economy

Delivery-Level Data

The delivery dataset contains:

- Delivery ID
- Match ID
- Innings
- Over number
- Ball number
- Batting team
- Bowling team
- Batter
- Bowler
- Non-striker
- Batter runs
- Extra runs
- Total runs
- Extras type
- Wicket
- Player out
- Dismissal kind
- Legal delivery indicator

The data was transformed into analytical views before being used in Power BI.

7. Dashboard Design

The CricVision platform consists of **three main dashboard pages**.

Dashboard 1: Executive Dashboard

The Executive Dashboard provides an overall summary of the selected IPL season.

Main components:

KPI Cards

- Total runs
- Total wickets
- Other important season-level performance indicators

Champion and Runner-up

Displays the champion and runner-up teams for the selected season.

Orange Cap

Displays the leading run scorer of the selected season along with the player's runs and dynamic player image.

Purple Cap

Displays the leading wicket-taker of the selected season along with wickets and dynamic player image.

Top Fours and Sixes

Displays the leading players based on fours and sixes.

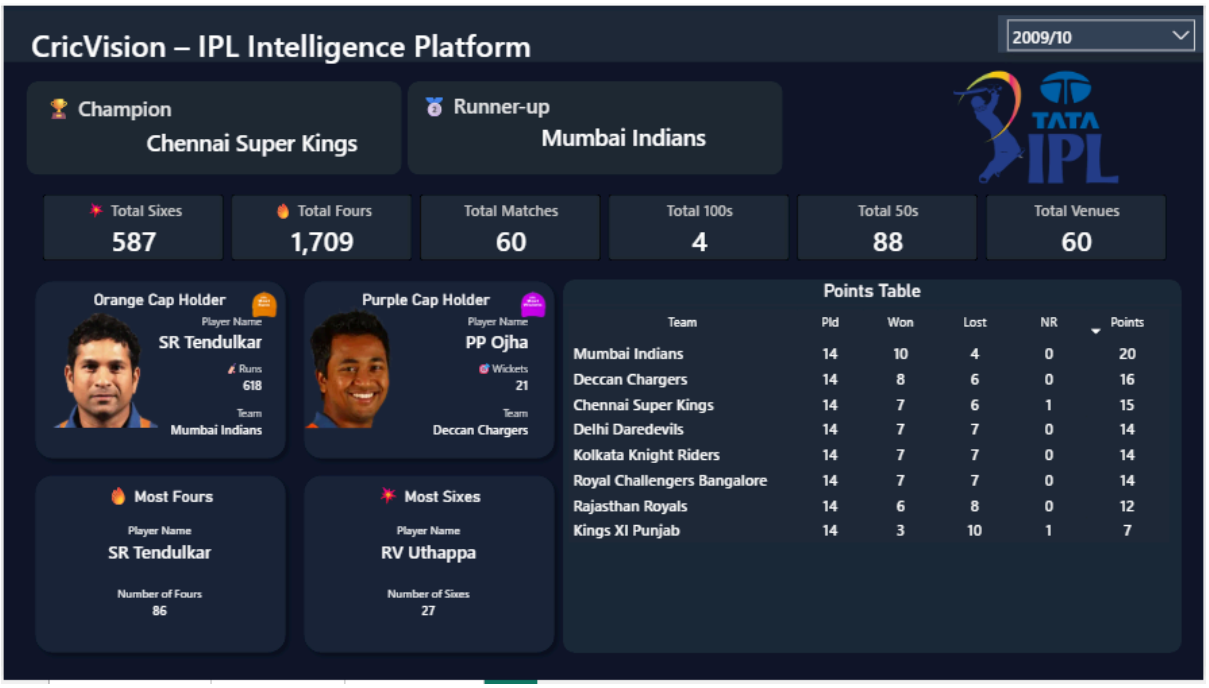
Points Table

Displays: Team, Matches played, Wins, Losses, No results, Points

The points table is calculated using league-stage matches while excluding playoff matches.

Season Slicer

Allows users to select a particular IPL season and dynamically update the dashboard.



Dashboard 2: Batting Intelligence

The Batting Intelligence page focuses on individual batting performance.

Components:

Player Slicer

Allows users to select a particular batsman.

Batting KPIs: Total runs, Batting average, Strike rate, Fours, Sixes, Half-centuries, Centuries

Player Profile

Displays: Selected player, Team, Matches played, Dynamic player image

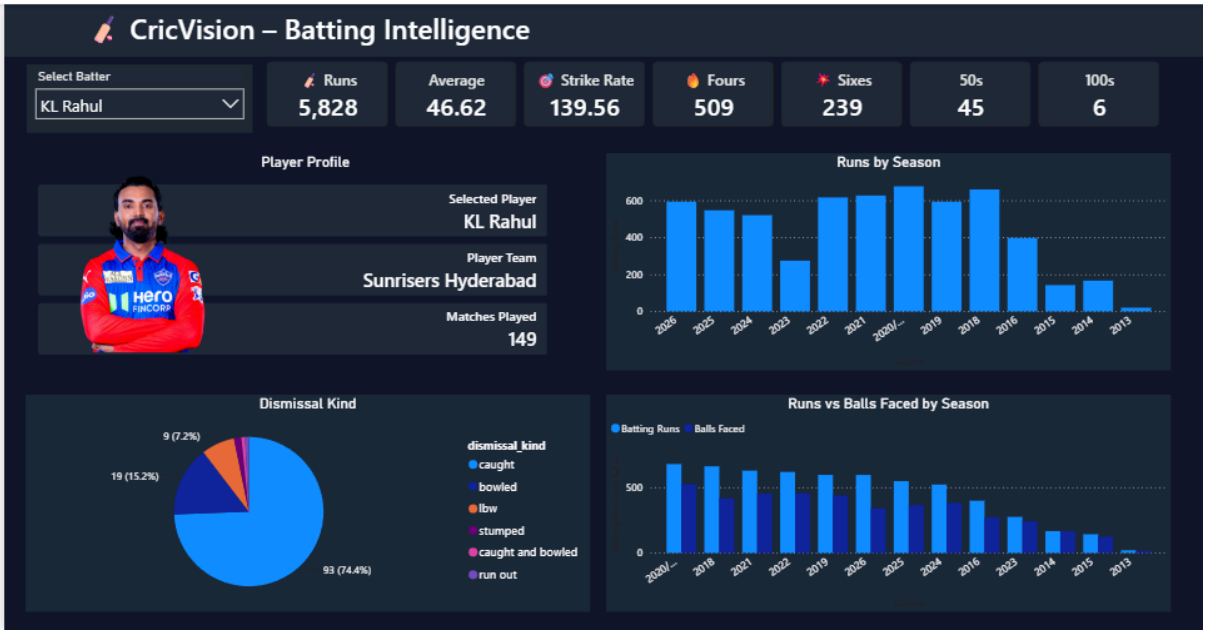
Runs by Season

Shows the player's run-scoring performance across seasons.

Batting Performance Charts

Additional charts provide insights into the player's batting performance and season-wise statistics.

The page allows users to select different players and immediately observe changes in their statistics.



Dashboard 3: Bowling Intelligence

The Bowling Intelligence page focuses on individual bowling performance.

Components:

Bowler Slicer

Allows users to select a particular bowler.

Bowling KPIs

- Total wickets
- Bowling average
- Economy
- Bowling strike rate
- Dot balls
- Five-wicket hauls
- Best bowling figures

Bowler Profile

Displays:

- Selected bowler
- Team

- Matches played
- Dynamic player image

Wickets by Season

Shows the selected bowler's wicket-taking performance across seasons.

Top Batters Dismissed

Displays the batsmen dismissed most frequently by the selected bowler.

Wickets Against Teams

Shows the teams against which the selected bowler has taken the most wickets.

The dashboard uses interactive filtering to allow users to explore bowling performance across different seasons and players.



8. Key Insights

The dashboard enables several important insights to be derived from IPL data.

8.1. Player Performance Varies Across Seasons

The Batting Intelligence page allows users to compare a player's runs, average, strike rate, fours, and sixes across different seasons.

8.2. Different Players Demonstrate Different Batting Strengths

Some players contribute through consistent run scoring, while others generate greater impact through boundaries and sixes.

8.3. Bowling Performance Can Be Analysed Beyond Wickets

The Bowling Intelligence page combines wickets with economy, strike rate, dot balls, and bowling average to provide a broader understanding of bowling effectiveness.

8.4. Bowler Matchups Can Be Identified

The Top Batters Dismissed analysis helps identify batsmen against whom a particular bowler has been successful.

8.5. Team Standings Provide Season-Level Performance

The points table allows users to compare teams based on matches played, wins, losses, no results, and points.

8.6. Major Season Achievements Can Be Identified Quickly

The Executive Dashboard provides quick access to the Champion, Runner-up, Orange Cap, and Purple Cap for the selected season.

8.7. Interactive Filtering Improves Analysis

Season and player slicers allow users to explore specific areas of the dataset without manually changing individual charts.

9. Challenges Faced

Several challenges were encountered during the development of the project.

9.1. Large and Complex Dataset

The IPL dataset contains a large number of match and delivery-level records, making data organization and querying important.

9.2. Different Table Structures

Batting, bowling, match, player, and delivery information were stored separately and required appropriate joins and relationships.

9.3. Season Filtering

Ensuring that KPIs and visuals responded correctly to the selected season required careful modelling and DAX implementation.

9.4. Player Name Matching

Player names from different datasets sometimes followed different naming conventions. A common calculated player dimension was created to improve consistency.

9.5. Points Table Calculation

Playoff matches needed to be excluded from league-stage points calculations. SQL logic was developed to identify and exclude the final four playoff matches from each season.

9.6. Dynamic Player Images

Player images had to be matched with the exact player names used in the Power BI model so that images could change dynamically with the player slicer.

9.7. Power BI Relationship Management

Creating correct relationships between dimensions, analytical views, and source tables was necessary to ensure slicers filtered the visuals correctly.

9.8. DAX Calculation Issues

Some measures required additional logic to ensure that career-level statistics were not displayed when a season-specific result was expected.

10. Real Life Applications

CricVision can be useful for different stakeholders in the sports industry.

Cricket Analysts

Analysts can use the dashboard to study player performance, identify trends, and compare players across seasons.

Team Management

Team management can analyse batting and bowling strengths and identify areas requiring improvement.

Coaches

Coaches can study player performance and identify successful player-versus-player matchups.

Players

Players can use performance statistics to understand their strengths, weaknesses, and consistency.

Sports Journalists

Journalists can use historical statistics to support match previews, articles, and performance analysis.

Fantasy Cricket Analysts

Player statistics such as runs, wickets, strike rate, economy, and consistency can help in fantasy-cricket analysis.

Sports Data Analysts

The project provides a foundation that can be extended into predictive analytics and machine learning applications.

11. Conclusion

The **CricVision: Interactive IPL Cricket Analytics Dashboard** demonstrates how raw cricket data can be transformed into meaningful and interactive sports intelligence.

The project combines **MySQL, SQL, Power BI, and DAX** to collect, process, analyse, and visualize IPL data. SQL views and queries were used to structure the data, while DAX measures were used to create dynamic performance indicators.

The three dashboard pages provide different levels of analysis, from an overall season summary to detailed batting and bowling performance.

The use of interactive slicers, KPI cards, performance charts, points tables, player profiles, and dynamic player images makes the platform easy to explore and understand.

Overall, the project demonstrates the application of data analytics and visualization techniques to the sports domain and provides a foundation for more advanced cricket analytics solutions.

12. Future Scope

The CricVision platform can be extended in several ways.

12.1. Player Comparison

Develop a dedicated comparison page where two or more players can be compared based on batting and bowling statistics.

12.2. Predictive Analytics

Machine learning models can be introduced to predict:

- Match winners
- Player performance
- Expected runs
- Expected wickets

12.3. Win Probability

A real-time win-probability model could be developed using current match conditions and historical data.

12.4. Head-to-Head Analysis

The platform could analyse detailed batter-versus-bowler matchups.

12.5. Advanced Player Metrics

Additional metrics such as:

- Boundary percentage
- Dot-ball percentage
- Strike-rate consistency
- Bowling impact
- Player consistency scores

could be introduced.

12.6. Real-Time IPL Data

The dashboard could be connected to a live cricket data source to provide real-time match analytics.

12.7. AI-Generated Match Insights

Generative AI could be integrated to automatically generate natural-language summaries of player and team performances.

12.8. Venue Analysis

A dedicated venue analytics module could analyse:

- Average scores
- Batting-friendly venues
- Bowling-friendly venues
- Team win percentages
- Toss impact

12.9. Advanced Team Analytics

Team performance could be analysed based on powerplay, middle overs, death overs, batting partnerships, and bowling phases.

12.10. Mobile Dashboard

The dashboard could be optimized and published for mobile devices so that users can access IPL insights from anywhere.

13. References

1. **Cricsheet** – Structured cricket data and IPL match datasets.
2. **Indian Premier League (IPL)** – Official tournament information and statistics.
3. **MySQL Documentation** – SQL queries, joins, aggregation, views, and window functions.
4. **Microsoft Power BI Documentation** – Data modelling, Power Query, DAX, slicers, and visualizations.
5. **Microsoft DAX Documentation** – Measures, calculated tables, filtering, and aggregation functions.